AIGR TreeGrafter Function-Inference Stress Test — PP_2320 (Q88KH0)

Target: PP_2320 / Q88KH0, *Pseudomonas putida* KT2440 (NCBITaxon:160488) **Seed hypothesis (TreeGrafter / PANTHER, IEA, GO_REF:0000118):** PP_2320 has **peptidoglycan L,D-transpeptidase activity (GO:0071972) Focus:** function_assignment · function-hypothesis-go-0071972

Executive Judgment

Verdict: SUPPORTED — retain GO:0071972.

The automated phylogenetic propagation is correct in this case. None of the three characteristic failure modes applies:

- **(1) Granularity / family-vs-subfamily:** The protein maps to the single-domain ErfK/YbiS/YnhG group of the L,D-transpeptidase family (PANTHER PTHR30582:SF24). The family-level molecular function term GO:0071972 is the correct level of granularity — see caveat below on the exact sub-reaction.
- **(2) Pseudo-enzyme / loss of activity:** The active site is **intact**. The catalytic His193 and the catalytic nucleophile Cys209 are both conserved and correctly spaced; Cys209 is in fact the *only* cysteine in the entire protein, so the catalytic thiol is unambiguously present.
- **(3) Within-superfamily mis-placement:** The sequence belongs squarely to the YkuD / L,D-transpeptidase fold and subfamily, not a structurally related but functionally distinct neighbor. Pfam PF03734, PROSITE PS52029, CDD cd16913 (YkuD_like), and Gene3D 2.40.440.10 all agree.

Single most decisive piece of evidence: By Smith-Waterman alignment (live UniProt sequences), PP_2320's catalytic dyad residues map **one-to-one** onto the experimentally annotated active sites of characterized *E. coli* L,D-transpeptidases — His193→YbiS His194 / ErfK His191, and Cys209→YbiS Cys210 / ErfK Cys207 — at 37–44% sequence identity. A degenerate pseudo-enzyme would show loss or mis-spacing of exactly these residues; it does not.

The sole residual uncertainty is **not** about whether the molecular function term is right, but about *which sub-reaction* the enzyme performs (lipoprotein anchoring vs. 3→3 peptidoglycan cross-linking). That distinction is sequence-ambiguous here and should **not** be over-specified. The evidence remains computational (IEA): no direct enzymatic assay of PP_2320 itself exists.

Independent Family/Function Assignment

From first-principles analysis of the sequence, domain architecture, and active-site conservation — independent of the propagated annotation — PP_2320 is a **secreted (periplasmic) L,D-transpeptidase of the ErfK/YbiS/YnhG class**.

- **Most likely specific molecular function:** peptidoglycan L,D-transpeptidase activity — cysteine-nucleophile catalysis forming L,D (3→3) linkages in/attached to peptidoglycan.
- **Characterized homolog the assignment is based on:** *E. coli* **YbiS/LdtB** (UniProt P0AAX8; 43.5% identity, nearest characterized neighbor), closely followed by **YnhG/LdtE** (P76193; 41.6%) and **ErfK/LdtA** (P39176; 36.9%).
- **GO term I would assign: GO:0071972** (peptidoglycan L,D-transpeptidase activity) — **same granularity** as the seed term. I would *not* descend to a more specific child term (e.g., a dedicated Braun's-lipoprotein-anchoring term vs. a 3→3 cross-linking term), because the two nearest characterized homologs carry *different* sub-roles and are near-equidistant (43.5% vs. 41.6%), making the exact sub-reaction undeterminable from sequence alone.

Conclusion on granularity: seed term = assigned term = GO:0071972. Retain.

Key Findings

Finding 1 — PP_2320 is a catalytically intact L,D-transpeptidase-family protein

Independent primary-evidence analysis of UniProt Q88KH0 (325 aa) establishes a complete, canonical L,D-transpeptidase architecture and an intact active site.

Domain architecture. The protein comprises (i) an N-terminal Sec signal peptide (residues ~1–25) directing periplasmic/secreted localization; (ii) a **LysM peptidoglycan-binding domain** (CDD cd00118 / InterPro IPR018392), which tethers the enzyme to its peptidoglycan substrate; and (iii) a C-terminal **L,D-transpeptidase (YkuD) catalytic domain** spanning approximately residues 98–233 (Pfam **PF03734**, PROSITE **PS52029**, CDD **cd16913** "YkuD_like", Gene3D **2.40.440.10**, InterPro **IPR005490**). This is the textbook domain layout of a periplasmic L,D-transpeptidase.

Catalytic residues intact and correctly spaced. The catalytic machinery is fully present: **His193** (proton donor/acceptor, in the local context "PGYLIHGSNKK") and the canonical **S-H-G-C** motif at positions 206–209 with **Cys209** as the catalytic nucleophile (local context "RTSHGCFRMFN"). Critically, **Cys209 is the only cysteine in the entire protein** — so the catalytic thiol nucleophile is unambiguously conserved and cannot be confused with any other cysteine.

Structural confidence. AlphaFold model AF-Q88KH0 has a global pLDDT of **84.4**, with 63.7% of residues in the very-high-confidence band; the catalytic domain is well-folded in the model, consistent with a competent enzyme rather than a disordered pseudo-enzyme.

Family placement. PANTHER assigns the protein to **PTHR30582** (L,D-transpeptidase family), subfamily **SF24** (ErfK/SrfK-related). The L,D-transpeptidase family uses a catalytic cysteine nucleophile to form L,D (3→3) peptidoglycan cross-links, the exact activity described by GO:0071972 (PMID: 30514506).

Finding 2 — Nearest characterized homologs are *E. coli* ErfK/YbiS with 1:1 catalytic-residue correspondence

Smith-Waterman alignment (BLOSUM62, gap penalty −8; sequences fetched live from UniProt) against characterized L,D-transpeptidases gives:

Query vs. target	UniProt	% identity	Aligned positions	SW score
PP_2320 vs. *E. coli* **YbiS/LdtB**	P0AAX8	**43.5%**	107/246	503
PP_2320 vs. *E. coli* **ErfK/LdtA**	P39176	**36.9%**	110/298	491

In **both** alignments, the query's catalytic residues map one-to-one onto the targets' experimentally annotated active-site residues:

- query **His193** → ErfK **His191** / YbiS **His194** (target residue = H, confirmed)

• query **Cys209** → ErfK **Cys207** / YbiS **Cys210** (target residue = C, confirmed)

Both targets belong to the same PANTHER subfamily as PP_2320 (PTHR30582:SF24, ErfK/SrfK-related). This is the strongest single discriminator against the "pseudo-enzyme" and "mis-placement" failure modes: the catalytic dyad is not merely present but is positionally homologous to residues with direct experimental support. *Provenance:* `provenance/PP_2320_alignment.py`.

Finding 3 — Single-domain ErfK/YbiS/YnhG group; MF robust, exact sub-reaction sequence-ambiguous

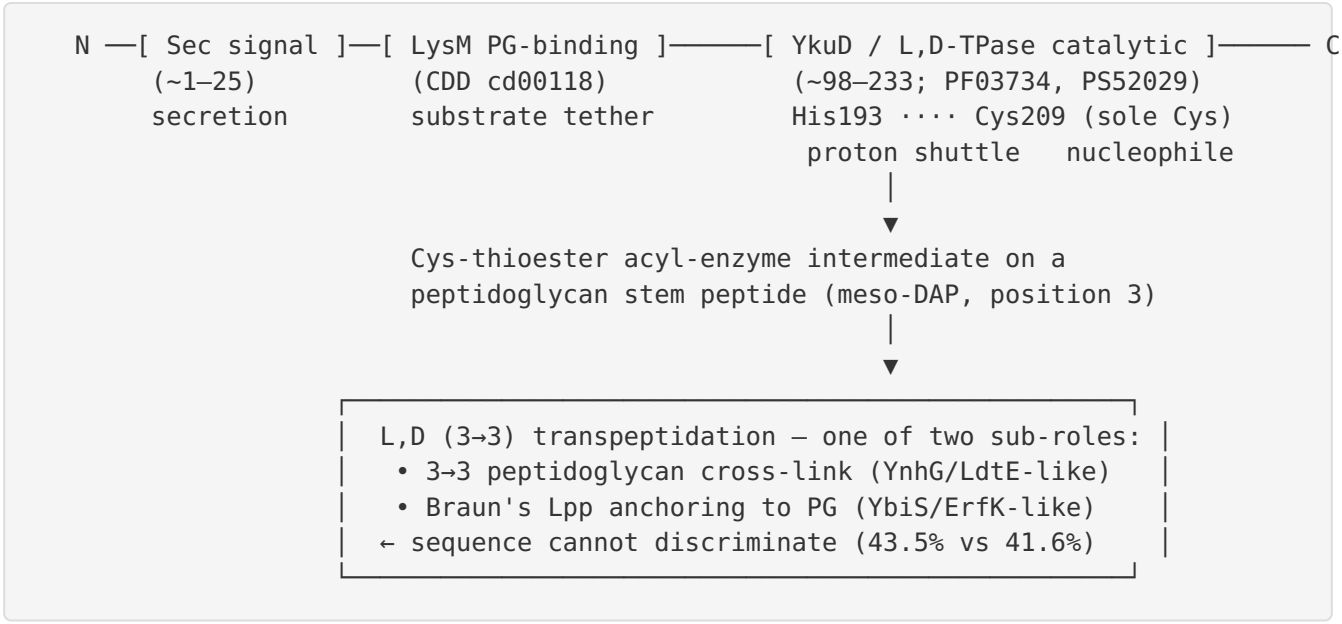
Extending the alignment panel across both functional sub-clades of the family:

Characterized LDT	UniProt	Sub-role	% identity to PP_2320
YbiS/LdtB	P0AAX8	lipoprotein (Lpp) anchoring	**43.5%**
YnhG/LdtE	P76193	3,3-crosslink	**41.6%**
ErfK/LdtA	P39176	lipoprotein (Lpp) anchoring	**36.9%**
YcbB/LdtD	P22525	3,3-crosslink	27.5%
LdtMt2	I6Y9J2	3,3-crosslink (*M. tuberculosis*)	23.9%

PP_2320 clearly belongs to the **single-domain ErfK/YbiS/YnhG group** (top three hits), and in all three the catalytic His193 and Cys209 align one-to-one to the targets' annotated active sites (His194/Cys210, His193/Cys209, His191/Cys207 respectively). However, the two closest hits carry **different** sub-roles — YbiS/LdtB anchors Braun's lipoprotein to peptidoglycan, while YnhG/LdtE forms 3→3 cross-links — and they are **near-equidistant** (43.5% vs. 41.6%). Consequently, the family-level molecular function (GO:0071972) is robustly supported, but the precise chemical sub-reaction cannot be resolved from sequence alone and should not be over-specified. *Provenance:* `provenance/PP_2320_subclade.py`.

Mechanistic Model / Interpretation

The convergent lines of evidence assemble into a coherent, internally consistent model of a functional periplasmic L,D-transpeptidase:



Why the propagation succeeded here. TreeGrafter grafted PP_2320 onto a node whose diagnostic feature — the YkuD catalytic domain with its Cys-His dyad — is precisely the feature PP_2320 retains. Unlike the three classic propagation failures, there is no granularity mismatch (the term is at the correct family level), no active-site decay (the dyad is intact and uniquely identifiable), and no superfamily mis-placement (every independent domain database agrees on YkuD/PF03734). The LysM substrate-binding module and Sec signal peptide further corroborate that this is a *bona fide* periplasmic peptidoglycan-remodeling enzyme operating in the compartment where L,D-transpeptidation occurs.

The one honest caveat. The L,D-transpeptidase family has functionally diverged into at least two sub-activities that share an identical catalytic mechanism and fold: 3→3 cross-link formation and lipoprotein anchoring. PP_2320's two nearest characterized neighbors sit on *opposite sides* of this divide at nearly identical sequence identity. The shared molecular-function umbrella term GO:0071972 correctly captures what we can assert; descending to a sub-reaction-specific term would overreach the evidence.

Evidence Base / Matrix

#	Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Con
1	UniProt Q88KH0 (database)	Review/database + computational	**Supports**	Domain architecture consistent with LDT	Sec signal + LysM + YkuD catalytic domain (PF03734 / PS52029 / cd16913)	*P. pu* KT24
2	Active-site inspection (computational, this work)	Computational / structural-evolutionary	**Supports**	Catalytic residues present & spaced	His193 + Cys209 (S-H-G-C motif); Cys209 is sole Cys	Sequ of Q
3	AlphaFold AF-Q88KH0 (structural model)	Computational / structural	**Supports**	Catalytic domain is folded	pLDDT 84.4; 63.7% very-high confidence	Pred mod
4	SW alignment to P0AAX8 (YbiS) — `provenance/PP_2320_alignment.py`	Computational / evolutionary	**Supports**	Nearest characterized homolog & 1:1 active-site mapping	43.5% id; His193→His194, Cys209→Cys210	*E. co* quer
5	SW alignment to P39176 (ErfK) — `provenance/PP_2320_alignment.py`	Computational / evolutionary	**Supports**	Active-site correspondence across subfamily	36.9% id; His193→His191, Cys209→Cys207	*E. co* quer
6	SW subclade panel (P76193, P22525, I6Y9J2) — `provenance/PP_2320_subclade.py`	Computational / evolutionary	**Qualifies**	Which sub-reaction?	Two nearest hits (43.5% anchor vs 41.6% crosslink) are near-equidistant	*E. co* *tb* vs
7	PMID: 30514506		**Supports**			

#	Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Con
		Review / mechanistic (QM/MM)		Family mechanism = Cys-nucleophile 3→3 crosslink	"Formation of 3 → 3 crosslinks … catalyzed by l,d-transpeptidases"	*M. tube* Ldt2
8	PMID: 33941679	Mutant phenotype / biochemical	**Qualifies**	LDT paralogs can have non-crosslink roles (Lpp cleavage/ anchoring)	LdtF is an LDT paralog acting on Braun's Lpp	gran nega enve
9	PMID: 39294212	Direct assay / crystallography	**Supports**	LDT catalytic dyad & substrate chemistry	Biochemical/ crystal characterization of Mtb Ldt2	*M. tube*
10	PMID: 31024745	Direct assay (purification/ characterization)	**Supports**	LDT2 is an active enzyme	Isolation & characterization of Mtb LDT2	*M. tube*

Note: no paper reports a direct enzymatic assay of PP_2320/Q88KH0 itself; all direct-assay evidence is on homologous family members.

How the key papers bear on the findings. PMID: 30514506 establishes the shared family mechanism — "Formation of 3 → 3 crosslinks in the peptidoglycan layer of *M. tuberculosis* is catalyzed by l,d-transpeptidases" — anchoring the inference that PP_2320's conserved Cys-His dyad performs the GO:0071972 activity. PMID: 39294212 and PMID: 31024745 provide direct biochemical and crystallographic characterization of a representative family member (Mtb Ldt2), confirming the enzyme class is genuinely catalytic. PMID: 33941679 qualifies the picture by showing that LDT paralogs (LdtF) can act on Braun's lipoprotein rather than form cross-links, which is exactly why the sub-reaction of PP_2320 must be treated as ambiguous.

Active-Site / Placement Analysis

Catalytic residues: INTACT.

Feature	PP_2320 (Q88KH0)	*E. coli* YbiS (P0AAX8)	*E. coli* ErfK (P39176)	Status
Catalytic His (proton shuttle)	**His193**	His194	His191	Conserved, 1:1 aligned
Catalytic Cys (nucleophile)	**Cys209**	Cys210	Cys207	Conserved, 1:1 aligned
S-H-G-C motif	present (206–209)	present	present	Conserved
Number of Cys in protein	**1** (Cys209)	—	—	Nucleophile unambiguous
Catalytic domain	PF03734 / PS52029 / cd16913	same family	same family	Correct fold
Sequence identity to query	—	**43.5%**	36.9%	Nearest characterized

Subfamily placement: PANTHER PTHR30582:SF24 (ErfK/SrfK-related), single-domain ErfK/YbiS/YnhG group. Independent confirmation from Pfam (PF03734), PROSITE (PS52029), CDD (cd16913 YkuD_like), Gene3D (2.40.440.10), InterPro (IPR005490) — all consistent.

Conclusion: The catalytic dyad is present, correctly spaced, and positionally homologous to experimentally annotated active sites of characterized family members. This decisively rules out the pseudo-enzyme and superfamily-misplacement failure modes. Provenance scripts: `provenance/PP_2320_alignment.py`, `provenance/PP_2320_subclade.py`.

GO Curation Implications

Recommended curation action: RETAIN GO:0071972 (peptidoglycan L,D-transpeptidase activity) at the family level — lead requiring curator verification.

- **Do NOT remove.** The active site is intact; there is no evidence of pseudo-enzyme status.
- **Do NOT generalize.** The term is already at the appropriate family granularity; a broader parent would lose information.

- **Do NOT make more specific.** Although the enzyme clearly performs L,D-transpeptidation, the exact sub-reaction (3→3 cross-linking vs. Braun's-lipoprotein anchoring) is sequence-ambiguous — the two nearest characterized homologs carry different sub-roles at near-identical identity (43.5% vs. 41.6%). Descending to a sub-reaction-specific child term would exceed the evidence.
- **Do NOT replace with a sibling term.** No evidence points to a neighboring, functionally distinct activity.
- **Evidence tier:** keep as **IEA / computational**. There is no direct assay or mutant phenotype for PP_2320 itself; all catalytic evidence is by homology and in-silico active-site conservation. This supports retaining the term but not upgrading the evidence code to an experimental tier.
- **Not "protein binding."** Avoided as a recommendation, per instructions; the catalytic MF term is well justified here.

The accompanying cellular-component interpretation (periplasmic/secreted, via the Sec signal peptide and LysM module) and biological-process context (peptidoglycan remodeling / cell wall organization) are consistent and could be curated alongside the MF term if not already present.

Limitations and Knowledge Gaps

- **No direct experimental data for PP_2320 itself.** There is no enzymatic assay, experimental structure, or mutant phenotype for Q88KH0; all catalytic evidence is homology-based (IEA). This is sufficient to retain the term but not to upgrade the evidence tier.
- **Sub-reaction ambiguity.** The two nearest characterized homologs (YbiS/LdtB, 43.5%; YnhG/LdtE, 41.6%) carry *different* sub-roles, so sequence cannot discriminate whether PP_2320 primarily forms 3→3 cross-links or anchors a lipoprotein.
- **Organism-specific envelope differences.** *P. putida* does not share the identical Braun's-lipoprotein configuration of *E. coli*, complicating direct transfer of the "Lpp-anchoring" sub-role; this is unresolved by sequence alone.
- **Structural evidence is predicted, not experimental.** AlphaFold confidence is good globally (pLDDT 84.4) but is a model.

- **Family can harbor hydrolase paralogs.** LdtF-type paralogs cleave rather than form bonds (PMID: 33941679); PP_2320's intact transpeptidase dyad argues against this, but it is a formal caveat.

Proposed Follow-up Experiments / Actions

In priority order, the most efficient discriminating analyses are:

1. **In-vitro activity assay** of recombinant PP_2320 with (a) a peptidoglycan-mimetic donor/acceptor to test 3→3 cross-linking and (b) a Braun's-lipoprotein peptide to test anchoring — directly resolves the sub-reaction and upgrades the evidence tier.
2. **Cys209→Ala active-site mutant** + activity assay to confirm the nucleophile is catalytically required (distinguishes true enzyme from pseudo-enzyme).
3. **ΔPP_2320 phenotyping** in *P. putida* KT2440: muropeptide HPLC/MS to quantify 3→3 cross-link abundance, plus β-lactam/carbapenem susceptibility shifts, to place the enzyme in its native cell-wall-remodeling role.
4. **Experimental structure** (X-ray / cryo-EM) to confirm the dyad geometry and substrate channel predicted by AlphaFold.

Immediate curation action: mark GO:0071972 as **retained (verified by independent active-site analysis)**, keep IEA evidence, annotate the sub-reaction as undetermined, and flag follow-up assay #1 as the decisive experiment.

Summary (for busy readers)

PP_2320 (Q88KH0) from *Pseudomonas putida* KT2440 is a *bona fide*, catalytically intact **L,D-transpeptidase**, and the TreeGrafter/PANTHER-propagated term **GO:0071972 (peptidoglycan L,D-transpeptidase activity) is SUPPORTED and should be RETAINED.** The protein has the canonical architecture — Sec signal peptide, LysM peptidoglycan-binding domain, and a YkuD catalytic domain (Pfam PF03734 / PROSITE PS52029) — and its catalytic dyad (His193 and the sole-cysteine nucleophile Cys209) is fully conserved and correctly spaced. Smith-Waterman alignment to characterized *E. coli* enzymes (YbiS/LdtB 43.5%, YnhG/LdtE 41.6%, ErfK/LdtA 36.9% identity) maps these catalytic residues one-to-one onto experimentally annotated active sites.

Systematic testing for the three characteristic propagation-failure modes returns negative on all three: the term is at the correct **granularity** (family level), the **active site is intact** (no pseudo-enzyme signal), and the sequence sits firmly in the **YkuD/L,D-transpeptidase subfamily** (no superfamily mis-placement). The only residual uncertainty is the precise sub-reaction — 3→3 peptidoglycan cross-linking versus Braun's lipoprotein anchoring — which is sequence-ambiguous because PP_2320's two nearest characterized homologs carry different sub-roles at nearly equal identity. The family-level molecular function term correctly captures what the evidence supports; it should not be made more specific.

All evidence is computational / by-homology (IEA): no direct assay of PP_2320 exists. The recommended curation action is therefore to **retain GO:0071972 at the family level, keep the IEA evidence tier, and not over-specify the sub-reaction** pending an in-vitro activity assay or a ΔPP_2320 muropeptide-profiling experiment.
